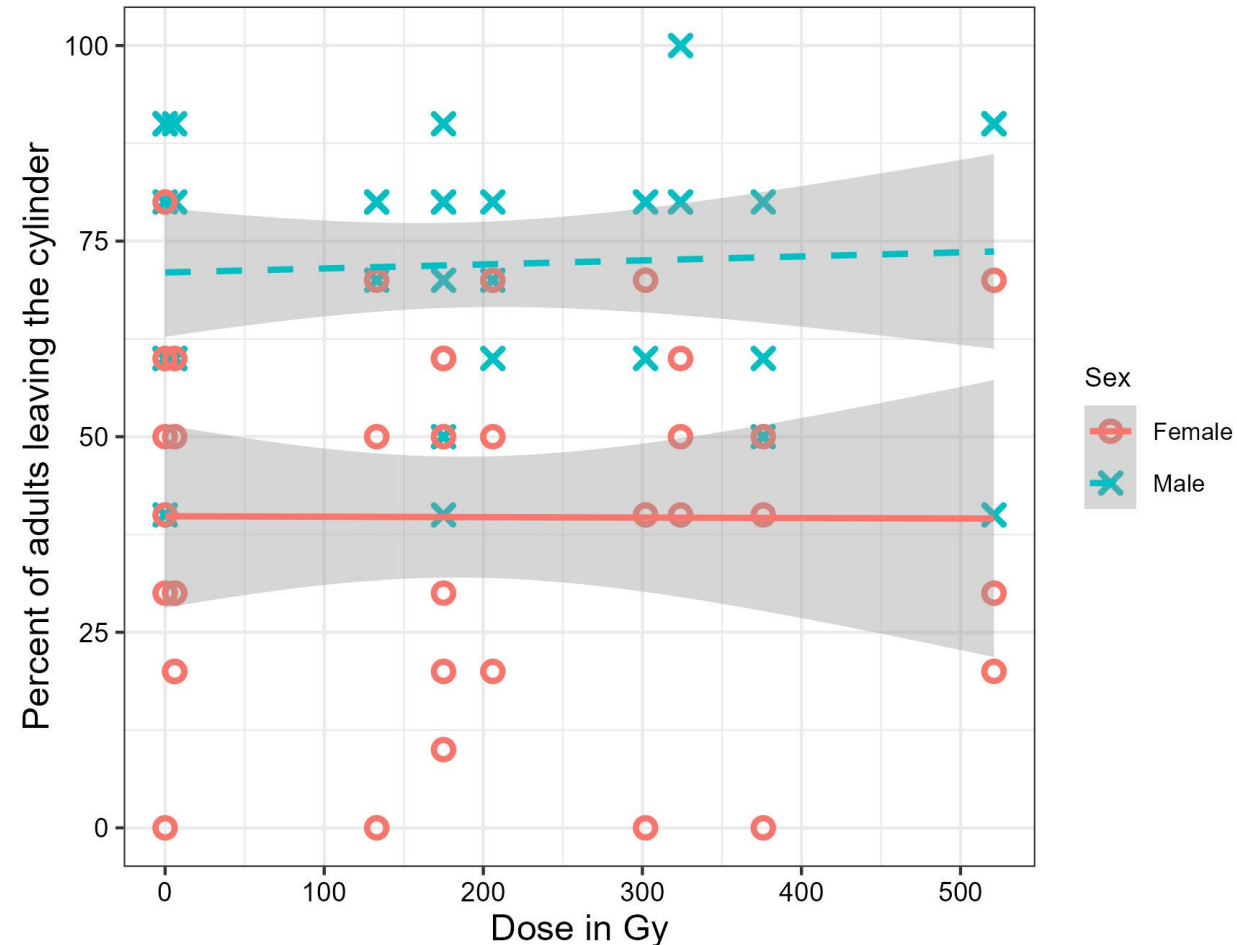


Experiment 1: Flight

GLM with quasibinomial distrib. residual df = 65

Effect	LR χ^2	df	P
Dose (Gy)	0.001	1	0.98
Sex	43.83	1	< 0.001
Run (Cohort)	2.38	1	0.12
Dose $\times$ Sex	0.061	1	0.80
Dose $\times$ Run	0.129	1	0.72
Sex $\times$ Run	0.018	1	0.89

Experimental unit: a cage of 10 moths

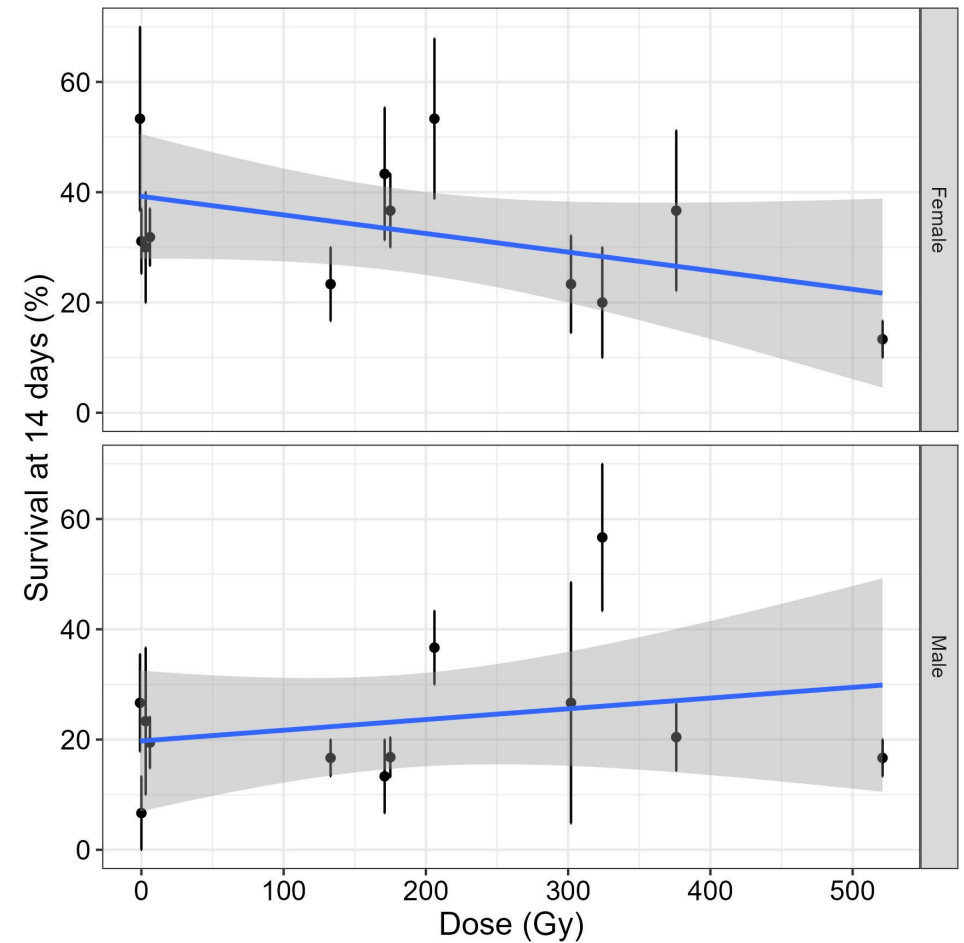


Experiment 1: Longevity

GLM with quasibinomial distribution, resid. df = 65

Effect	LR χ^2	df	<i>P</i>
Dose (Gy)	0.650	1	0.42
Sex	5.056	1	< 0.025
Run (Cohort)	3.761	1	0.052
Dose $\times$ Sex	3.870	1	0.049
Dose $\times$ Run	2.439	1	0.118
Sex $\times$ Run	0.674	1	0.412

Experimental unit: a cage of 10 moths

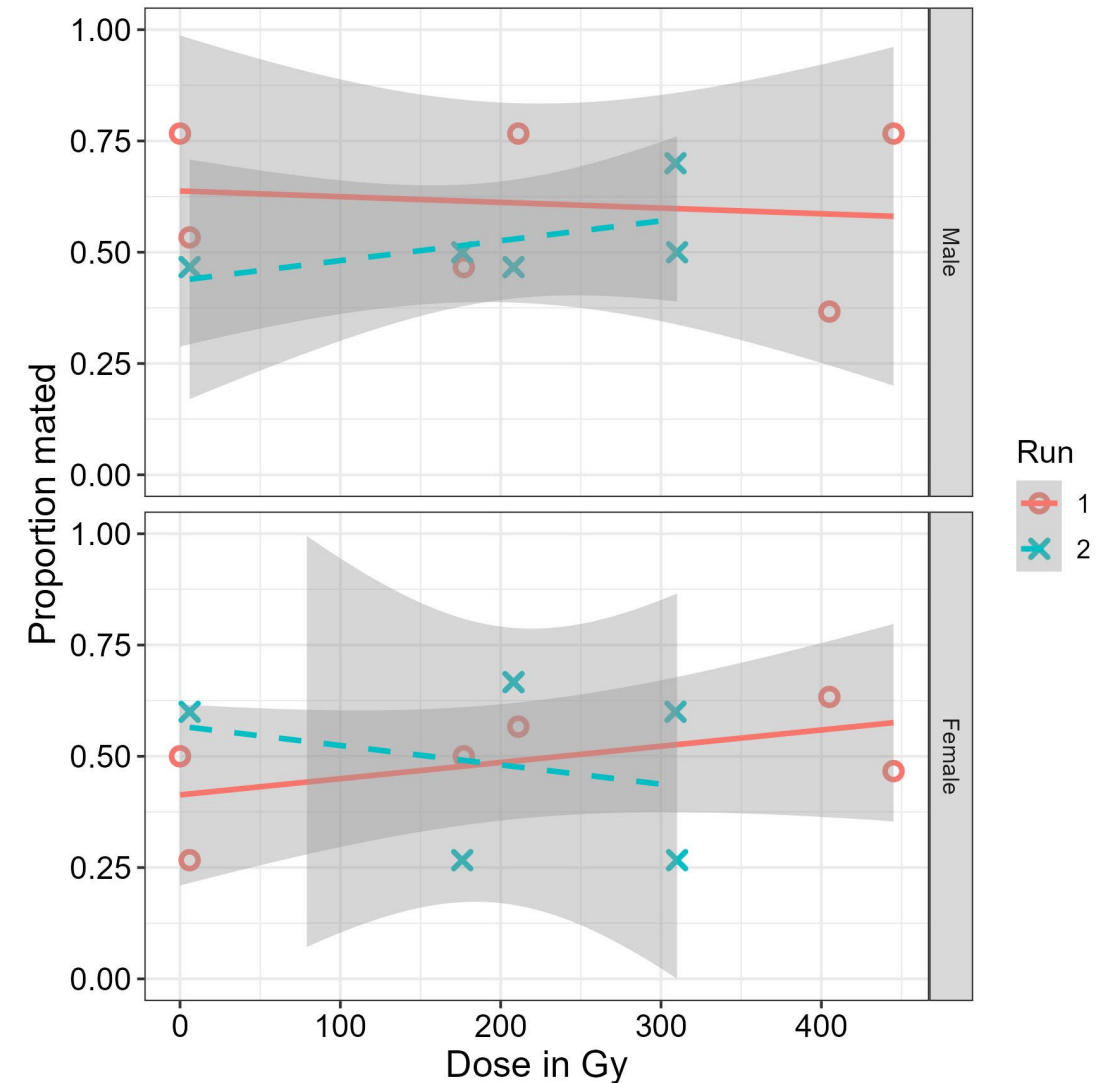


Experiment 2: Mating

GLM with binomial distribution, resid. df = 653

Effect	LR χ^2	df	P
Dose (Gy)	0.150	1	0.698
Sex	2.996	1	0.083
Run (Cohort)	0.576	1	0.448
Dose × Sex	0.271	1	0.602
Dose × Run	0.144	1	0.704
Sex × Run	0.988	1	0.320

Experimental unit: individual moth

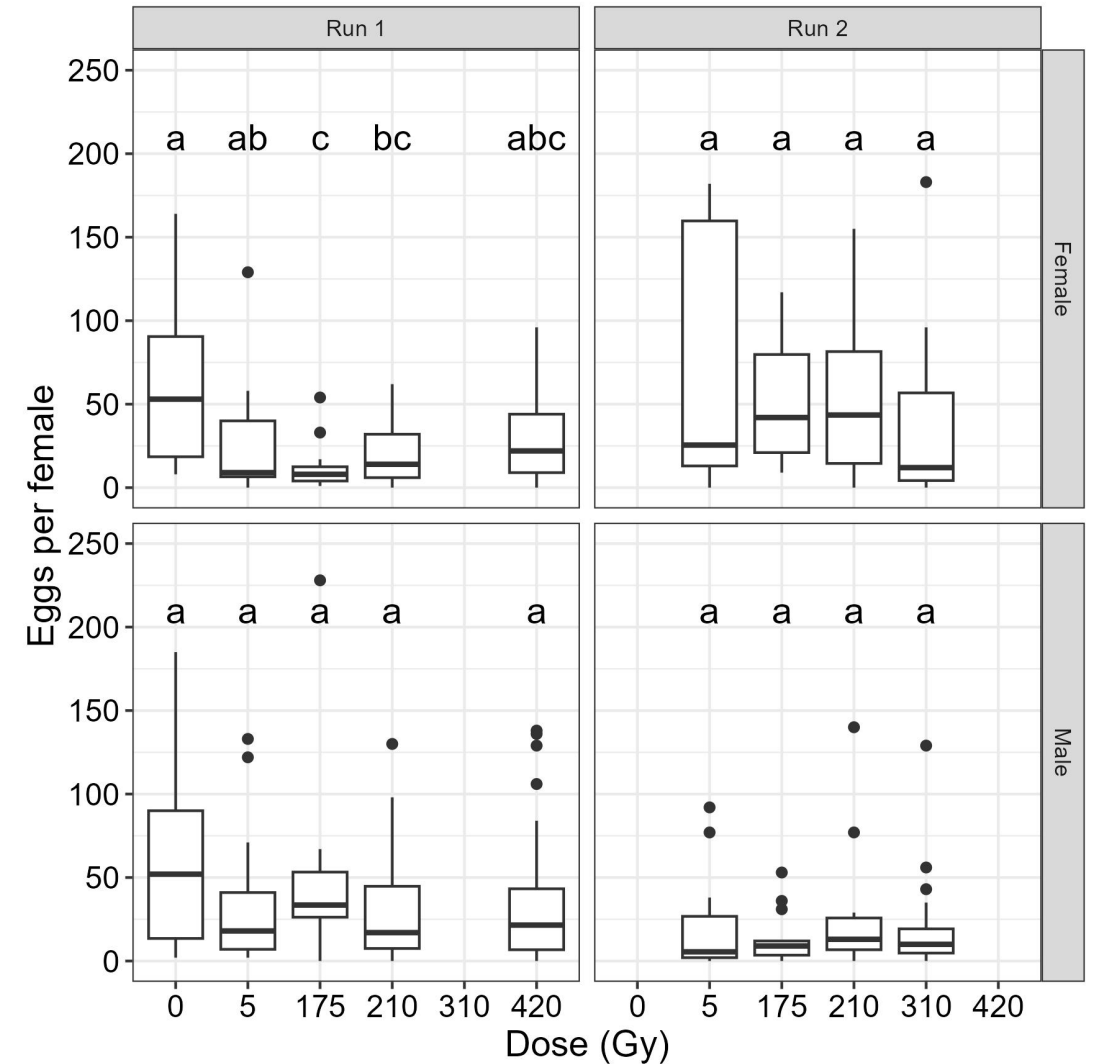


Experiment 2: Oviposition

GLM with negative binomial distribution, resid. df = 333

Effect	LR χ^2	df	P
Dose (Gy)	23.5	1	< 0.001
Sex	2.57	1	0.12
Run (Cohort)	0.92	1	0.34
Dose × Sex	5.73	5	0.33
Dose × Run	0.59	2	0.74
Sex × Run	21.3	1	< 0.001

Experimental unit: individual moth



Experiment 2: Development to pupa

GLM with zero-inflated nb distribution, resid. df = 331

Effect	Wald χ^2	df	P
Dose (Gy)	81.64	5	< 0.001
Sex	0.80	1	0.370
Run (Cohort)	5.87	1	0.015
Dose × Sex	9.71	5	0.084
Dose × Run	11.91	2	0.003
Sex × Run	20.44	1	< 0.001

Experimental unit: individual moth

